

PSEUDOCODE WRITING

← Back

QUESTIONS

Q1

Student Management System
[BL3 – Apply]
10 marks

Q2

Library Book Management [BL3 – Apply]
10 marks

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

Q5 5. Vehicle Registration System [BL3 – Apply]

10 marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:

- TwoWheeler — with engine capacity
- FourWheeler — with seating capacity and AC status

Apply inheritance to reuse common attributes and override a displayDetails() method in each subclass.

OOP Concepts: Inheritance | Method Overriding | Subclass

Your Answer

```
class Vehicle {  
    protected String regNo;  
    protected String owner;  
    protected String fuelType;  
  
    public Vehicle(String regNo, String owner, String fuelType) {  
        this.regNo = regNo;  
        this.owner = owner;  
        this.fuelType = fuelType;  
    }  
}
```

Q3 ✓
Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4 ✓
Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5 ✓
5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6 ✓
6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7 ✓
7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8 ✓
8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9 ✓
9. Library Management System
[BL4 – Analyze]
10 marks

Your Answer

```
public void displayDetails() {  
    System.out.println("Vehicle Registration Details:");  
    System.out.println("Reg No    : " + regNo);  
    System.out.println("Owner     : " + owner);  
    System.out.println("Fuel Type : " + fuelType);  
}  
}  
  
class TwoWheeler extends Vehicle {  
    private double engineCapacity;  
  
    public TwoWheeler(String regNo, String owner, String fuelType, double engineCapacity) {  
        super(regNo, owner, fuelType);  
        this.engineCapacity = engineCapacity;  
    }  
  
    @Override  
    public void displayDetails() {  
        super.displayDetails();  
        System.out.println("Engine Capacity: " + engineCapacity + " CC");  
        System.out.println("-----");  
    }  
}
```

Q3

Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks

Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks

Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks

Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

Your Answer

```
class FourWheeler extends Vehicle {
    private int seatingCapacity;
    private boolean acStatus;

    public FourWheeler(String regNo, String owner, String fuelType, int seatingCapacity, boolean acStatus) {
        super(regNo, owner, fuelType);
        this.seatingCapacity = seatingCapacity;
        this.acStatus = acStatus;
    }

    @Override
    public void displayDetails() {
        super.displayDetails();
        System.out.println("Seating Capacity: " + seatingCapacity + " persons");
        System.out.println("AC Status      : " + (acStatus ? "Available" : "Not Available"));
        System.out.println("-----");
    }
}

public class Main {
    public static void main(String[] args) {
        Vehicle bike = new TwoWheeler("TN09AB1234", "Karthik Valli", "Petrol", 150);
        Vehicle car = new FourWheeler("TN09CD5678", "Naga Tejaswitha", "Diesel", 5, true);
    }
}
```


Q4

Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks



Q5

5. Vehicle Registration System
[BL3 – Apply]
10 marks



Q6

6. Bank Account Hierarchy [BL4
– Analyze]
10 marks



Q7

7. Hospital Patient Record
System [BL4 – Analyze]
10 marks



Q8

8. E-Commerce Product Catalog
[BL4 – Analyze]
10 marks



Q9

9. University Staff Hierarchy
[BL4 – Analyze]
10 marks



Q10



```
this.seatingCapacity = seatingCapacity;  
this.acStatus = acStatus;  
}  
  
@Override  
public void displayDetails() {  
    super.displayDetails();  
    System.out.println("Seating Capacity: " + seatingCapacity + " persons");  
    System.out.println("AC Status      : " + (acStatus ? "Available" : "Not Available"));  
    System.out.println("-----");  
}  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Vehicle bike = new TwoWheeler("TN09AB1234", "Karthik Valli", "Petrol", 150);  
        Vehicle car = new FourWheeler("TN09CD5678", "Naga Tejaswitha", "Diesel", 5, true);  
  
        bike.displayDetails();  
        car.displayDetails();  
    }  
}
```

 Save Answer